Unit - 3

Solved proplems:

(2) The following are the number of hours 15 students spent studying.

2, 4, 3, 5, 2, 6, 4, 3, 5, 7, 2, 4, 6, 5, 3

Arrange the values

The unique values are

2, 3, 4, 5, 6, 7

Study hrs	frequency
2	3
3	3
4	3
5	3
6	2
7	1
Total	15

Calculate percentage

formula:

$$\text{percentage} = \frac{f}{N} \times 100$$

f = frequency

N = total no. of. observation = 15

hrs	frequency	percentage
2	3	$3/15 \times 100 = 20\%$
3	3	20%
4	3	20%
5	3	20%
6	2	$2/15 \times 100 = 13.33\%$
7	1	$1/15 \times 100 = 6.67\%$
total	15	100%

The highest frequency is 3, occurring for 2,3,4 and 5 hrs.

3.) Data distribution - Quartiles and IQR

consider the following dataset

10, 12, 15, 18, 20, 22, 25, 28, 30, 35

no. of observations:

$n = 10$

Find Q_2 (median)

since $n = 10$, the median is the avg of the 5th and 6th values.

$$Q_2 = \frac{20+22}{2}$$

$$\boxed{Q_2 = 21}$$

Find Q_1

lower half:

10, 12, 15, 18, 20

The middle value is

$$\boxed{Q_1 = 15}$$

Find Q_3

upper half.

22, 25, 28, 30, 35

$\boxed{Q_3 = 28}$

calculate IQR

formula:

$IQR = Q_3 - Q_1$

$IQR = 28 - 15$

$\boxed{IQR = 13}$

$Q_1 = 15,\ Q_2 = 21,\ Q_3 = 28,\ IQR = 13$

(4) Outlier Treatment - IQR method.

consider the dataset.

10, 12, 13, 14, 15, 16, 17, 18, 20, 50

Find Q1

10, 12, 13, 14, 15

$\therefore$

$\boxed{Q_1 = 13}$

Find Q_3

16, 17, 18, 20, 50

$\therefore$ $\boxed{Q_3 = 18}$

Calculate IQR

$IQR = Q_3 - Q_1$

$= 18 - 13$

$\boxed{IQR = 5}$

Calculate lower boundary

formula:

$L = Q_1 - 1.5(IQR)$

$= 13 - 1.5(5)$

$= 13 - 7.5$

$\boxed{L = 5.5}$

Calculate upper boundary

$U = Q_3 + 1.5(IQR)$

$= 18 + 1.5(5)$

$= 18 + 7.5$

$\boxed{U = 25.5}$

To identify the

< 5.5 or > 25.5

$50 > 25.5$

$\therefore$ 50 is an outlier

outlier treatment

Median $= \frac{15+16}{2} = 15.5$

50 can be replaced by 15.5

Outlier $= 50$

Treated data:

10, 12, 13, 14, 15, 15.5, 16, 17, 18, 20.

5) suppose the exam scores are 40, 42, 45, 46, 48, 50, 57, 90 detect the outlier using Z-score method

given:

$\mu = 51.625$

$\sigma = 15.09$

formula:

$$z = \frac{x - \mu}{\sigma}$$

15.69

$$z = \frac{28.375}{15.69}$$

$$\boxed{z \simeq 2.45}$$

Apply the rule

$|z| > 3$

$2.45 < 3$

$\therefore$ 90 is not an outlier according to the $\pm$ 3 rule

6 Consider 2, 3, 4, 4, 5, 6, 10 determine the type of skewness.

Given:

2, 3, 3, 4, 4, 5, 5, 6, 10

Calculate mean:

$$\text{Mean} = \frac{\Sigma x}{n}$$

$\Sigma x = 2+3+3+4+4+5+5+6+10 = 42$

$n = 9$

$$\text{mean} = \frac{42}{9}$$

mean = 4.67

Median

$\frac{9+1}{2} = 5$

5th values:

$\boxed{\text{median} = 4}$

Compax mean and median

mean = 4.67

median = 4

mean > median

the distribution is positively skewed

∴ Distribution is positively right skewed.

7. The waiting time for a bus is uniformly distributes between 0 and 20 minutes. Find the probability that a passenger waits between 5 and 12 minor.

soln: To find:

$P(5 < x < 12)$

formula:

$$P(c < x < d) = \frac{d-c}{b-a}$$

$\therefore a = 0, b = 20 / c = 5, d = 12$

Substitute,

$$P(5 < x < 12) = \frac{12-5}{20-0}$$

$$= \frac{7}{20}$$

$$= 0.35$$

convert to percentage.

$$0.35 \times 100 = 35\%$$

$$P(5 < x < 12) = 0.35$$

$$= 35\%$$

$$\boxed{35\%}$$

⑧. The heights of students are normally distributed with $\mu = 170$ cm, $\sigma = 10$cm, Find the z-score for a student whose height is 185 cm

Soln:-

Height are normally distributed with

$\mu = 170$cm

$\sigma = 10$ cm

Find the z-score of $x = 185$ cm.

formula :

$$z = \frac{x - \mu}{\sigma}$$

substitute

$$Z = \frac{180 - 170}{10}$$

$$\Rightarrow Z = \frac{15}{10}$$

$$\boxed{Z = 1.5}$$

Final ans:

$$\boxed{Z = 1.5}$$

The students height is 1.5 standard deviation above the mean.

(9) using the previous example, find the probility that a randomly selected student has a height less than 165cm?

soln:

from sol 8.

$z = 1.5$

use the standard of normalable.

for $z = 1.5$

$P(z < 1.5) = 0.9332$

Convert to percentage

$$0.9332 \times 100 = 93.32\%$$

Final ans:-

$$P(x < 185) = 0.9332$$

$$\boxed{93.32\%}$$

Interpretation:-

Approximately 93.32% of students are expected to have heights below 185 cm.

Verified